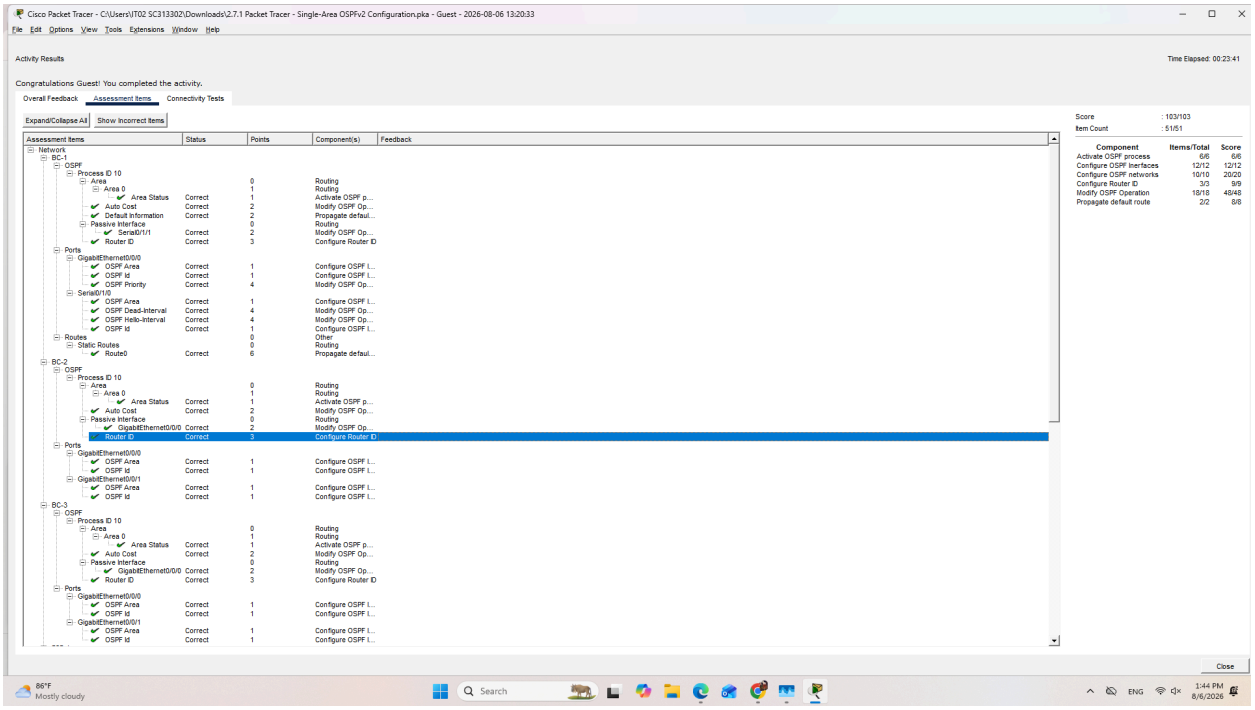
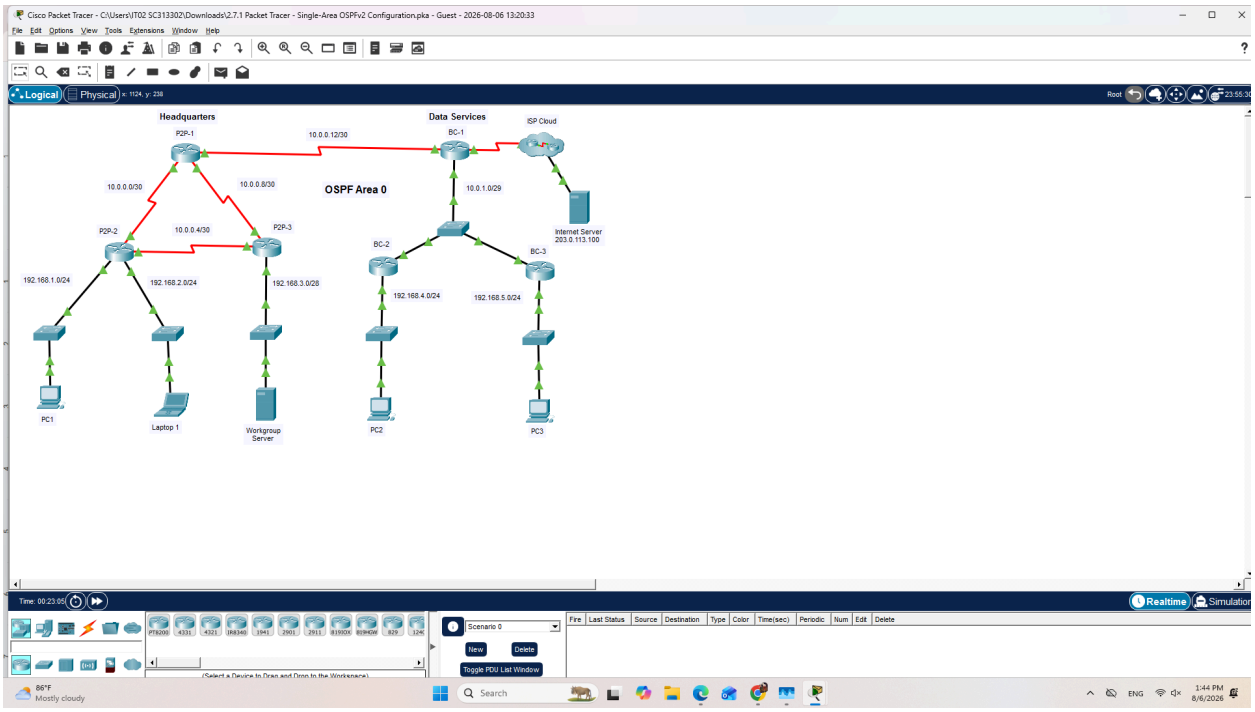

Cisco Packet Tracer - C:\Users\JT02 SC313302\Downloads\2.7.1 Packet Tracer - Single-Area OSPFv2 Configuration.pka - Guest - 2026-08-06 13:20:33
File Edit Options View Tools Extensions Window Help
Activity Results
Time Elapsed: 00:23:41
Congratulations Guest! You completed the activity.
Overall Feedback
Assessment Items
Connectivity Tests
Expand/Collapse All
Show Incorrect Items
Assessment Items
Status
Points
Component(s)
Feedback
Score : 103/103
Item Count : 51/51
Component
Items/Total
Score
Activate OSPF process 6/6 6/6
Configure OSPF Interfaces 12/12 12/12
Configure OSPF networks 10/10 20/20
Configure Router ID 3/3 9/9
Modify OSPF Operation 18/18 48/48
Propagate default route 2/2 8/8
Close
86°F
Mostly cloudy
Search
ENG
1:44 PM
8/6/2026


Cisco Packet Tracer - C:\Users\JT02 SC313302\Downloads\2.7.1 Packet Tracer - Single-Area OSPFv2 Configuration.pka - Guest - 2026-08-06 13:20:33
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
Headquarters
Data Services
ISP Cloud
P2P-1
BC-1
10.0.0.12/30
10.0.0.0/30
10.0.0.8/30
OSPF Area 0
10.0.1.0/29
Internet Server
203.0.113.100
P2P-2
10.0.0.4/30
P2P-3
BC-2
BC-3
192.168.1.0/24
192.168.2.0/24
192.168.3.0/28
192.168.4.0/24
192.168.5.0/24
PC1
Laptop 1
Workgroup Server
PC2
PC3
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
86°F
Mostly cloudy
Search
ENG
1:44 PM
8/6/2026

Topology

The network is made up of two segments joined by a single serial link. The Headquarters network (P2P-1, P2P-2, P2P-3) is a full-mesh of point-to-point serial links, activated with OSPF network statements and inverse masks. The Data Service network (BC-1, BC-2, BC-3) is a broadcast multiaccess Ethernet segment (10.0.1.0/29), activated per-interface with "ip ospf 10 area 0" instead of network statements. BC-1 bridges the two segments (link to P2P-1) and also connects out to the ISP cloud, where it holds the default route and originates it into OSPF for the rest of the network.

Addressing Table

Device	Interface	IP Address / Prefix
P2P-1	S0/1/0	10.0.0.1/30
P2P-1	S0/1/1	10.0.0.9/30
P2P-1	S0/2/0	10.0.0.13/30
P2P-2	S0/1/0	10.0.0.2/30
P2P-2	S0/1/1	10.0.0.5/30
P2P-2	G0/0/0	192.168.1.1/24
P2P-2	G0/0/1	192.168.2.1/24
P2P-3	S0/1/0	10.0.0.6/30
P2P-3	S0/1/1	10.0.0.10/30
P2P-3	G0/0/0	192.168.3.1/28
BC-1	S0/1/0	10.0.0.14/30
BC-1	S0/1/1	64.0.100.2/30
BC-1	G0/0/0	10.0.1.1/29
BC-2	G0/0/0	192.168.4.1/30
BC-2	G0/0/1	10.0.1.2/29
BC-3	G0/0/0	192.168.5.1/24
BC-3	G0/0/1	10.0.1.3/29
Internet Server	NIC	203.0.113.100/24
PC 1	NIC	192.168.1.10/24
Laptop 1	NIC	192.168.2.20/24
Workgroup Server	NIC	192.168.3.14/28
PC 2	NIC	192.168.4.40/24
PC 3	NIC	192.168.5.50/24

Objective

Implement single-area OSPFv2 in both point-to-point and broadcast multiaccess networks.

Background

You are helping a network engineer test an OSPF set up by building the network in the lab where you work. The devices are already interconnected and the interfaces are already addressed, with connectivity established within the local LANs. The task is to complete the OSPF configuration according to the requirements below so that, once finished, all hosts can ping the Internet server.